

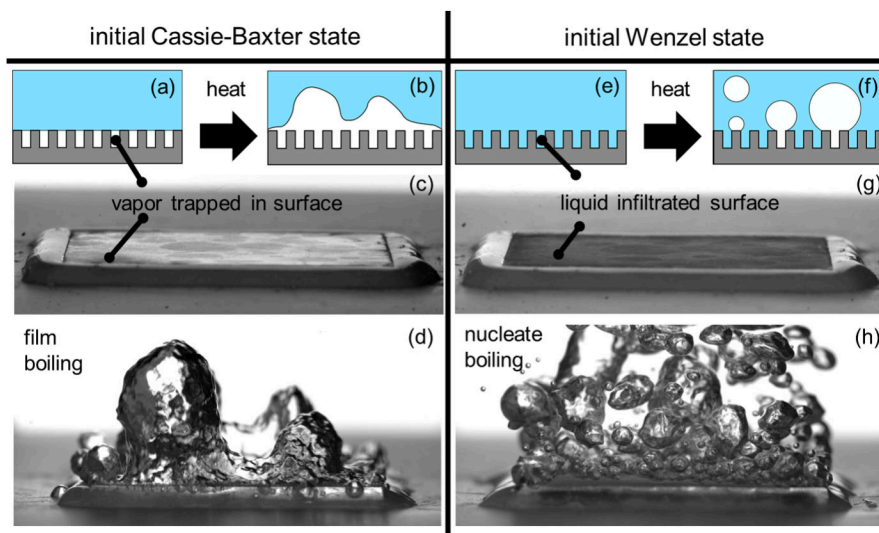


FIG. 2. Illustrations of (a) the initial Cassie-Baxter state progressing to (b) film boiling upon heating and (e) the initial Wenzel state progressing to (f) nucleate boiling upon heating. Visualization of (c) the initial Cassie-Baxter state and (d) subsequent film boiling from a submerged nanostructured superhydrophobic surface (heat flux 9 W/cm^2 , surface superheat 17.7°C). Visualization of (g) the initial Wenzel state and (h) subsequent nucleate boiling from a submerged nanostructured superhydrophobic surface (7.6 W/cm^2 , 7.4°C). High-speed visualizations of each case, over a range of heat fluxes, are available in Video S1 of the Supplemental Material [32].